



INTRODUCTION TO ALETHEIA

Aletheia, headquartered in Hong Kong, is Asia's leading independent research provider (IRP). We are a licensed entity, regulated by the authorities in HK, the US and the UK. *Institutional Investor* (II) ranked us as the No.1 IRP in 2022.

TAG (the Technology Advisory Group) is Aletheia's premier technology research division and focuses on global hardware/semiconductor, aiming to deliver differentiated, timely and value-added services.

Our research approach is based on long-term and sustainable themes, and a principles-based investment thesis, that is, rising content value, share gain, pricing power and monopoly in supply & demand as well as an integrated/holistic approach.

Our subscription-based business model is supported by a wide range of clients - SWFs, LOs, large platform HFs, L/S HFs, family offices and corporates across the globe.

Subscriptions to TAG's services are available through hard dollar or CSA arrangements. We have official relationships with CSA brokers BAML, JPM, UBS, GS, MS, Instinet, BTIG/Fubon, Cathay Securities and Sinolink Securities.

ALETHEIA/TAG RESEARCH TEAM

**We are rapidly expanding our team
and coverage in 2H26**



Warren Lau, Sr. Analyst

- ▶ 30 years' equity research and chip industry experience
- ▶ Research focus: AI, SPE & Memory
- ▶ B.Eng., M.Eng.



Skye Chen, Sr. Analyst

- ▶ 25 years' equity research & chip industry experience
- ▶ Research focus: AI, networking, OEM/ODM, & components.
- ▶ BA & Masters in Marketing.



George Chang, Sr. Analyst

- ▶ > 30 years' research experience
- ▶ Research focus: Power/PMIC, passive components & ABF substrate.
- ▶ BA (Econ.). Fluent in English, Mandarin & Japanese.



Stefan Chang, Sr. Analyst

- ▶ 20 years' research experience
- ▶ Research focus: AI, fabless & memory
- ▶ BA & MBA



Angus Lin, Analyst

- ▶ 10 years' research experience
- ▶ Research focus: AI, fabless & Adv packaging
- ▶ BA, M.Sc. in Finance.



Steven Liu, Analyst

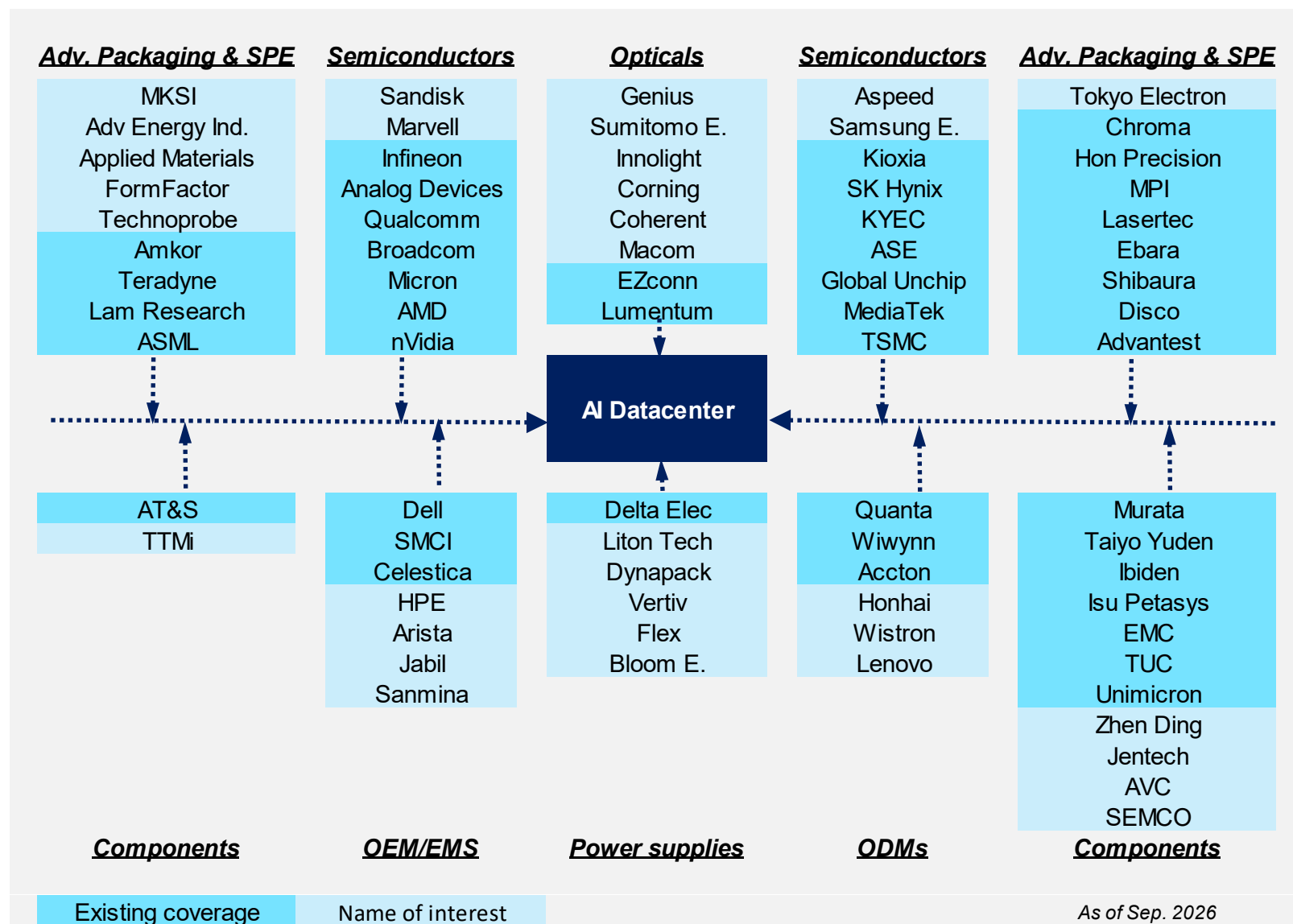
- ▶ 10 years' research experience
- ▶ Research focus: Adv material & packaging
- ▶ B.Eng./M.Eng./M.Sc. Fluent in English, Mandarin & Japanese.

RECENT RESEARCH FOCUS

- What could be the next bottleneck for the AI supply chain?
 - MLCC bookings continue to set all-time high
 - Reticle size of some GPUs/ASICs will reach 9-10x in 2H27/1H28 and could top 14-15x by 2H28/1H29
 - TDP for some XPU could top 8-12kW in 1H29 vs 4-5kW for GPUs in 2H27
 - PCB layers have surged to over 50 in 2H26
- Compute is no longer governed by a single GPU/ASIC or rack, but rather 'scaling up' multi-rack or pod to maximize system memory and bandwidth to meet the surging demand of multi-agentic workloads
- Spending on WFE and advanced packaging to accelerate into '27/'28, but supply may have a hard time playing catch-up
- Our best ideas: TSMC, Kioxia, Micron, Murata, AT&S, Lam Research, Lumentum, Advantest, Shibaura, MediaTek, MPI & Chroma

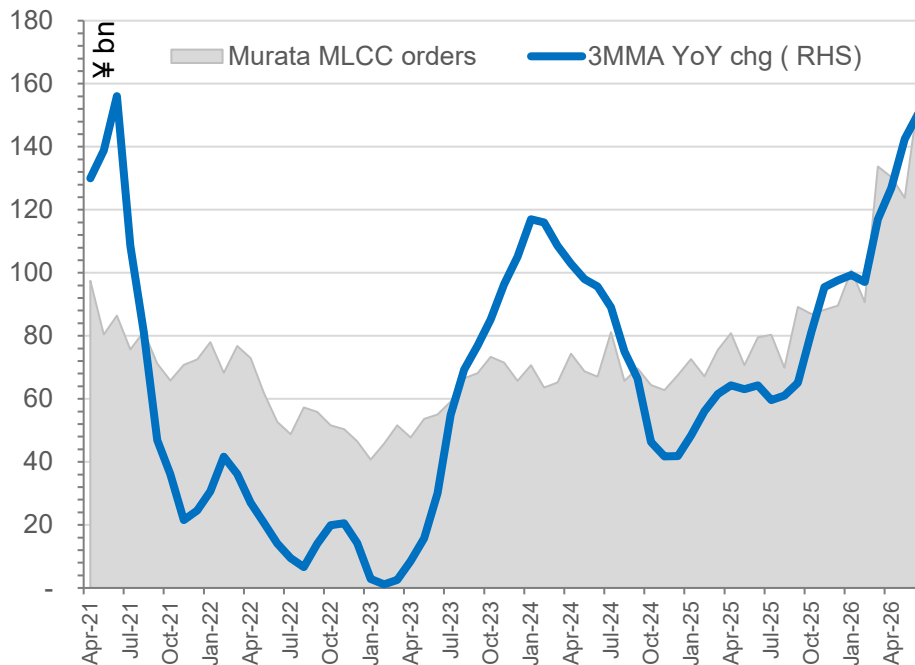


OUR COVERAGE OF AI/DC – AN HOLISTIC APPROACH



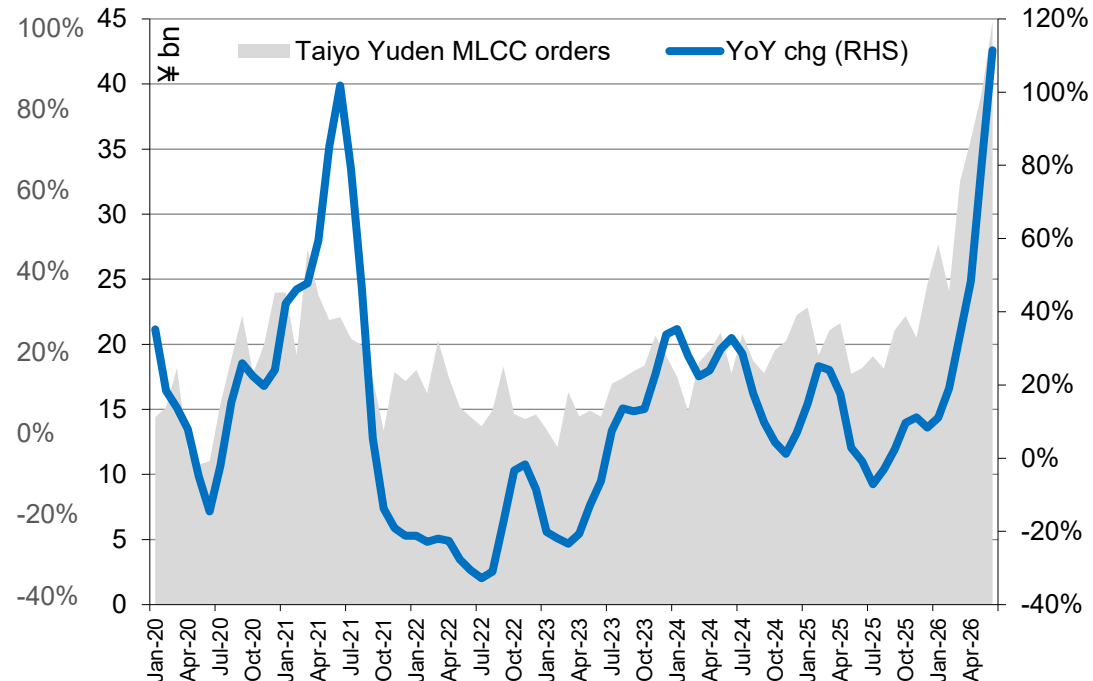
LATEST CHOKE POINT FOR AI SUPPLY CHAIN

Murata's Monthly Intakes of MLCC



Source: Company & Aletheia/TAG (Sep 2026)

Taiyo Yuden's Monthly Intakes of MLCC

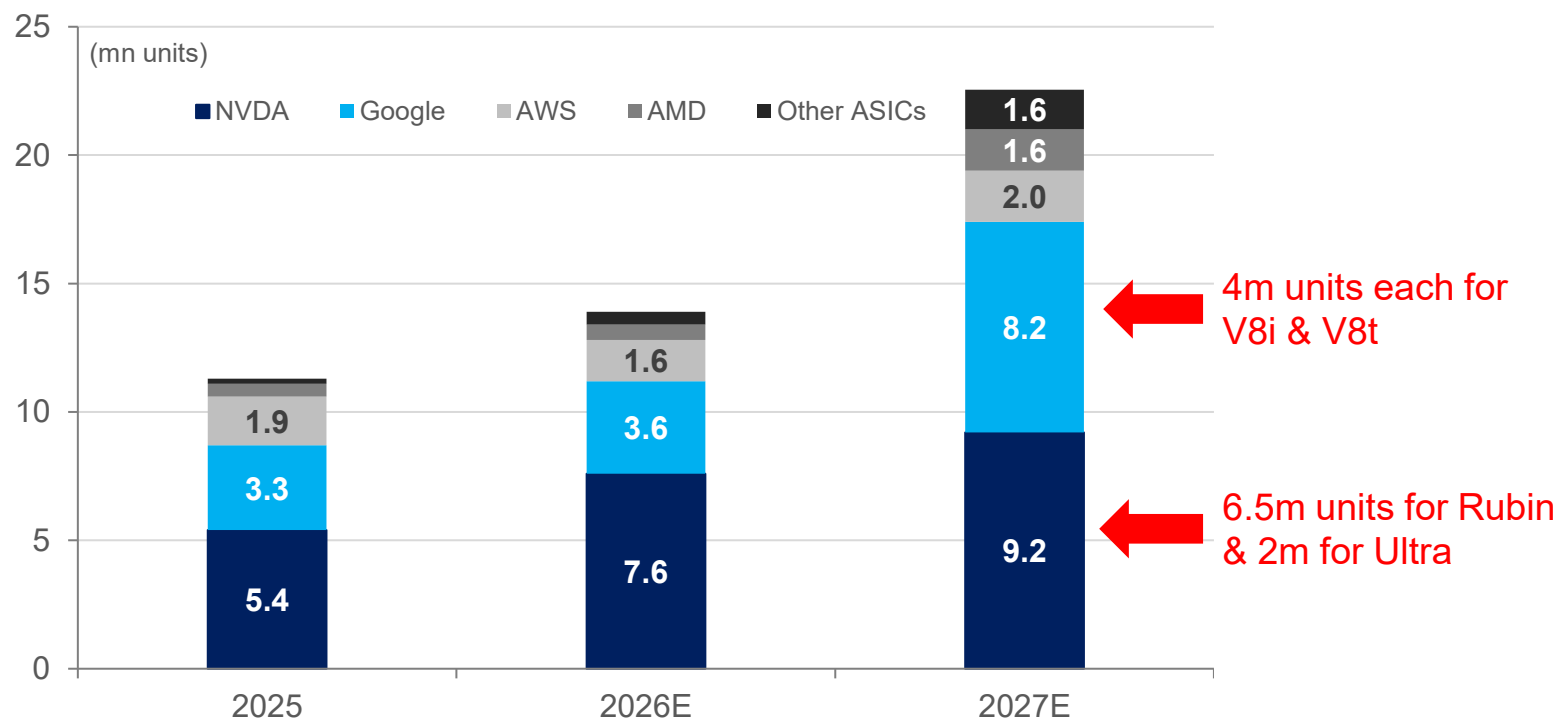


- ▶ July intakes almost doubled for Murata and were up 150% for Taiyo Yuden
- ▶ Latest BB ratio for Murata is 1.5x and Taiyo Yuden is >1.7x, way above the previous peak in COVID-19
- ▶ Consider this: TDP of TPU V9i is 7-8x of V8i and V10i is 2-3x of V9i

LATEST TRENDS AND UPDATES IN COMPUTE

- GPU/ASIC production to accelerate in 2027, with TPU and MI showing fastest pace while NVDA remains the top dog
- TPU and MI continue to exhibit aggressive spec upgrades while NVDA seems to be taking a pause (with Rubin Ultra)...
- ...but AI hardware design is shifting to multi-rack/pod (especially for multi-agentic AI workloads), emphasizing system density & bandwidth
- Server CPU productions to gain momentum into 2027, with more N2/3 capacity being available at TSMC
- Google's TPU supply may benefit the most from significant volume growth and surging content value gain

PRODUCTION OF AI DEVICES ACCELERATES IN '27



Source: Aletheia/TAG (Sep 2026)

- ▶ Overall units up 50-60% YoY in '27 led by TPU ASIC and MI GPU
- ▶ NVDA to remain the largest supplier in '27 but its share could dip to <50%
- ▶ Progress of the other ASICs remain slow but they aren't giving up

THE BULL CASE FOR NVDA

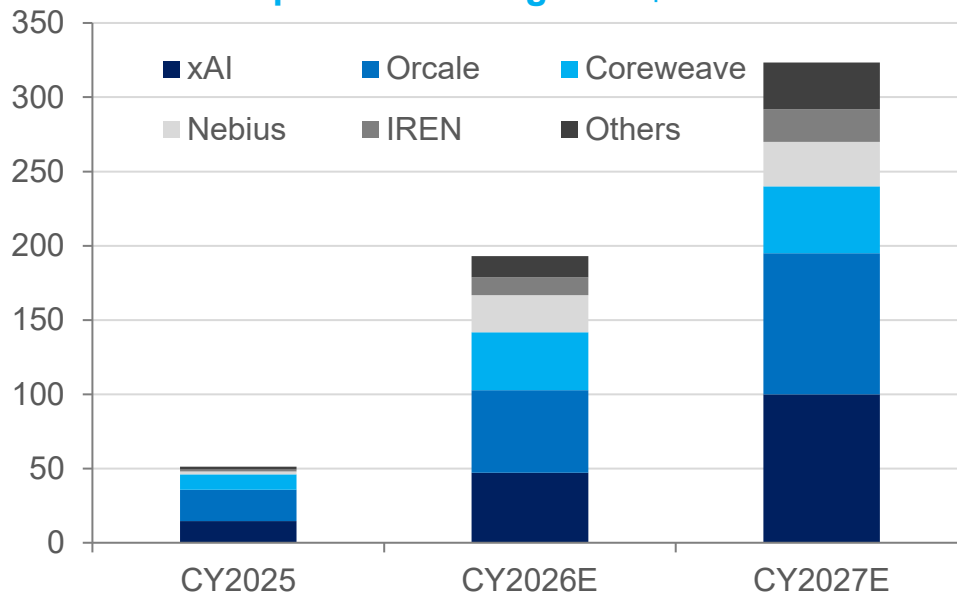
CoWoS (‘000 in pieces)	CY2025	CY2026E	CY2027E	Comments
H200	20	40	-	H200 shipments to China pending
B200/300	316	325	75	Extra orders from xAI and others in 3Q26 to be delivered in 1H27 (1.2-1.3m units)
Rubin	-	248	725	Expect 2.1m/6.5m unit production in 2H26 & 2027
Rubin Ultra	-	-	225	Tape-out in Oct-26 and HVM in 2H27 (2m)
Vera	-	125	270	AMKR and TSMC are the key suppliers. Expected 2.5/6m units in 2H26/27.
Total	336	738	1,295	These estimates exclude networking devices

Source: Aletheia/TAG (as of Sep 2026)

- Overall CoWoS production in 2027 suggests an increase of over 75% YoY but the value is likely to be considerably higher
- NVDA’s forecast of 70% revenue growth in FY28 is likely to be understated. We estimate YoY growth of at least 85%.

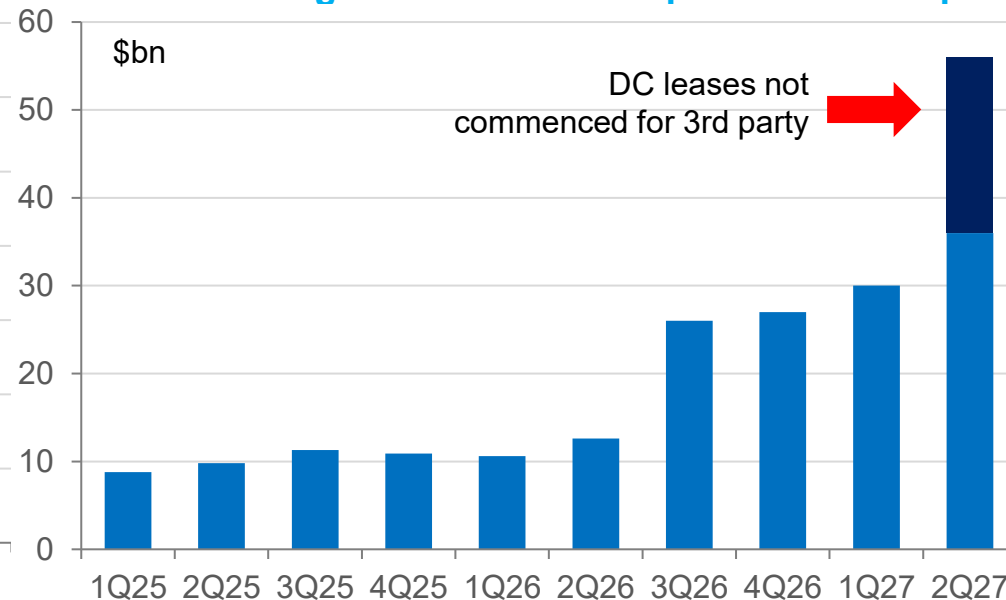
NVDA – FACTORS TO CONSIDER

Neo-Cloud Capex Could Surge to >\$300bn in '27



Source: Companies & Aletheia/TAG (Sep 2026)

Cloud Service Agreements - AI Compute Partnership*



Note: * Leaseback commitment to neo-cloud providers such as CRWV, NBIUS, IREN, Lambda, NSCALE, Firmus Tech, Oracle etc

Source: NVDA's 10-Q (Sep 2026)

► NVDA's total exposure to OpenAI to-date is estimated to be \$750bn (\$140bn direct & \$610bn indirect* exposure) and 12GW of compute.

► In July '26, NVDA reported the lengthiest DOI, AR days, cash conversion cycle and the lowest OP cash flow margin since CY23 (while revenue double YoY) – the beginning of AI era

Note: * indirect exposure includes OpenAI's cloud contract agreements with MSFT, ORCL & CRWV etc but excludes AWS which uses Trainium ASIC hardware.

AI HARDWARE DESIGN IS CHANGING

→ Optimized for agentic workloads

	Rubin	Rubin Ultra ¹ (Dual-die)	MI500 ³ (Quad-die)	TPU V8i	TPU V9i	OpenAI Jalapeno
Logic	2xN3	2xN3	4xN2+2xN3	2xN3P	4xN2P+4xN3P	1xN3P
I/O	1xN3+1xN4	1xN3+1xN4	4xN3	1xN3E	4xN3P	1xN3E
HBM	288GB/4	192GB/4 ² 256GB/4E	512GB/4E ³	288GB/3E	384GB/4E	216GB/4
Bandwidth	21TB/s	21/32TB/s	Up to 66TB/s	10-15TB/s	Up to 49TB/s	15.4TB/s
TDP	2.3kW	2.6kW	4.5-5kW	600W	4-5kW	700W
FP4	50PFLOPs	<50PFLOPs	~100PFLOPs	10.1PFLOPs	N.A.	13.4PFLOPs
HVM	2H26	2H27	4Q27	4Q26	1H28	4Q26

Note: 1-There is a quad-die Rubin Ultra SKU. 2-The final spec of Rubin Ultra is yet to be determined. At the time of writing, there are three options consisting of 192GB of HBM4 (8-Hi), 256GB HBM4E (8-Hi) and 386GB HBM4E (12-Hi). 3-There is a dual-die (with 12-Hi) MI500 GPU likely to go to Helios rack. The quad-die GPU is equipped with 12-Hi HBM4E.

Source: Aletheia/TAG (Sep 2026)

AI HARDWARE DESIGN IS CHANGING

Optimized for agentic workloads

	Rubin	Rubin Ultra ² (Dual-die)	MI500 ³ (Quad-die)	TPU V8i	TPU V9i	OpenAI Jalapeno
Scalability	576 ¹	576 ¹	512	1,152	1,600 ⁴	2,048
Networking	Spectrum-X	Spectrum-X	T100 or S1 G300	ICI	ICI	TH6
System compute	29EFLOPs	<29EFLOPs	51EFLOPs	12EFLOPs	N.A.	27EFLOPs
System memory	166TB ⁵ , 12PB/s	111-147TB ⁵ , 12-18PB/s	262TB, 34PB/s	332TB, 17PB/s	614TB, 78PB/s	432TB, 32PB/s
Device/tray Tray/rack	4/18	4/18	1/16	4/8	N.A.	8/16
Rack/pod	8	8	32	36	N.A.	16
TDP/rack	225kW	260kW	~100kW	<80kW	N.A.	~125kW
Rack HVM	1H27	1H28	1H28	Mid-27	2H28	2H27

Note: 1- We assume 8 racks as a pod but NVL Super pod can support as many as 16 racks or 1,152 GPUs. 2-There is a quad-die RU SKU. 3-The final spec of AMD's super pod structure is yet to be finalized. There is a dual-die MI500 GPU (each comes with 8 stacks of 12-Hi HBM4E) likely to go to Helios rack. 4-estimate based on TPU V8i. 5-VR200 rack comes with 576TB of high performance eSSD or 4,608TB of eSSD with eight racks pod. The eSSD density for Rubin Ultra is expected to rise further. Source: Aletheia/TAG (Sep 2026)

PROTOTYPE OF SCALE-UP 576 GPUS RACK POD -POLYPHE



Source: NVDA (2026 GTC), 2026 IEEE VLSI Symposium (June 2026)

STRUCTURE OF NVL SUPERPOD (40 RACKS)

Devices	VR NVL72 rack [18+9+2 trays]	Vera CPU rack [32 trays]	Groq LPX rack [32 trays]	BF4 ³ STX (storage) rack [16 trays]	Spectrum-6 SPX (networking) rack [4 trays]
GPU	72x	-	-	-	-
CPU	36x	256x	64x	32x	-
BF4 DPU	18x	-	-	64x	-
LPU	-	-	256x	-	-
NVLink	18x	-	-	-	-
SP6	-	-	-	-	16x
Density (GB)	VR NVL72 rack	Vera CPU rack	Groq LPX rack	BF4 STX (storage) rack	Spectrum-6 SPX (networking) rack
HBM	20,736	-	-	-	-
SoCAAM/DRAM	29,952 ²	196,608 ²	1,024 ²	20,544 ²	-
eSSD	576,000 ⁴	-	-	1,152,000 ⁴	-
SRAM	100 ¹	131 ¹	147	18 ¹	3 ¹
NVL SuperPod	VR NVL72 rack	Vera CPU rack	Groq LPX rack	BF4 STX (storage) rack	Spectrum-6 SPX (networking) rack
No of racks	16x	2x	10x	8x	4x
Total Density (TB)	VR NVL72 rack	Vera CPU rack	Groq LPX rack	BF4 STX (storage) rack	Spectrum-6 SPX (networking) rack
HBM	332	-	-	-	-
DRAM	479	393	10	164	-
SSD	9,216	-	-	9,216	-
SRAM	1.59 ¹	0.26 ¹	1.48	0.15 ¹	0.10 ¹

1- embedded SRAM (L1-L3) in logic/ASIC. The memory content value doesn't include this segment.

2-assume 768GB of SoCAAM/DRAM for VR and Vera racks, 386GB for BF4 STX storage rack and 16GB for LPX rack (the CPU can be x86 or Vera).

3-BF4 DSP uses the same compute core as Grace CPU.

4-In-rack (VR200 NVL72) eSSD or scalable eSSD in STX stock rack could use SLC NAND (still under qualification) or high-performance TCL NAND.

Source: nVidia (2026 GTC Taiwan) and Aletheia/TAG (Sep 2026)

SUPPLIES OF SERVER CPUs – THE RACE IS ON!

	2025	2026E	2027E	Remarks
nVidia	2.2m	~5.2m	6m	NVDA targets 6m units of Vera CPU production in '27, up from ~2.5m in 2H26
AMD	6.5m	up to 9m	up to 15m	Recently secured N2/3 wafers from TSMC for 2027 production. Working on CoWoS supply.
Intel	14m	~16m	~21m	Diamond Rapids delayed (twice) to '27. Clearwater Forest to be shipped from mid-'26.
Graviton	1.8m	~2.5m	~3.5m	Graviton 5 is a competitive device (192 cores on TSMC's N3)
Axion	<0.1m	~1m	~4.5m	Google to use Axion2 in the TPU V8 rack, replacing Intel's CPU. For agentic AI, Google still focuses on x86 CPUs (~85%).
Total	24.7m	33.7m	50m	CPU volume is expected to double in three years (among top-five suppliers), with AMD and Google growing at the fastest pace.

Source: Aletheia/TAG (June 2026)



PRODUCTIONS OF AMD'S VENICE & NVDA'S VERA

AMD-CoWoS	2026E	2027E	Remarks
TSMC	-	100k	Secured sufficient N2/N3 capacity and additional CoWoS-L capacity for server CPUs
ASE/SPIL	25k	200k	Acquired two legacy panel fabs to support AMD's server and consumer CPUs
AMKR	-	40k	HVM starts in mid-2027
Total	25k	240k	AMD's total CoWoS in 2027 could amount to 500-600k pieces, making it the fastest growing company globally
CPU units	400k	5m	Actual shipment depends yield loss, cycle times and reticle sizes

NVDA-CoWoS	2026E	2027E	Remarks
TSMC	40k	150k	TSMC adds AP7- P2 for CoWoS capacity to support Vera CPU
ASE/SPIL	-	-	No plan in 2026/27. They focus to serve AMD.
AMKR	70k	120k	Majority of AMKR's CoWoS-R supports Vera productions
Total	110k	270k	-
CPU units	2.5m	6m	Actual shipment likely to be less due to yield loss and cycle times.

Source: Aletheia/TAG (Sep 2026)

AMD's OPTIONALITY ENSURES ITS LEAD IN CPUs

	AMD				nVidia	Intel
Variants	Venice-Dense (6c)	Venice (Zen 6)	Verano (Zen 6)	Venice-X (Zen 6)	Vera ¹	Diamond Rapids ²
HVM	2H26	1H27	2H27	2H27	2H26	Mid-27?
Node	N2	N2	N2	N2	N3	18AP
Core/threads	256/512	128/256	72/144	96/192	88/176	256/-
Speeds	4GHz	5GHz	5GHz	5.2GHz	3.6GHz	4.5GHz
L3 cache	1,024MB	512MB	1,152MB (3D cache)	1,152MB (3D cache)	1644MB	1,280MB
Memory channels	16	16	24	16	8	16
Max DRAM density	8TB	8TB	4TB	8TB	1.5TB	8TB
Memory type	RDDIMN	RDDIMM	SoCAAM2	MRDIMM	SoCAAM2	R/MDIMM
Memory bandwidth	1.02TB/s	1.02TB/s	1.84TB/s	1.64TB/s	1.23TB/s	1.64TB/s
TDP	600W	500W	400W	600W	450W	600W
Workloads	Cloud Native/ agentic AI	General purpose	Low power/ local agentic AI	HPC	Low power/ agentic AI	HPC/ cloud Native

Note: 1-There will be an enhanced Vera CPU paired with Rubin Ultra GPU in 2H27.

2-Diamond Rapids has been delayed twice and the latest commercialization date has been pushed out to 2027.

Source: Aletheia/TAG (Sep 2026)

Densest memory (SRAM & LPDDR5X) per core!

THE BULL CASE FOR AMD

Estimated quarterly revenue/EPS by division

(bn)	4Q25	4Q26E	FY26E	FY27E	FY28E	Remarks
MI GPU	\$2.8	\$7.0	\$14.7	\$50.0	\$70.0	'27E estimate excludes potential Anthropic wins
Server CPU	\$2.6	\$5.0	\$17.7	\$39.5	\$68.5	Upside to '27&'28 volume as AMD has rescued more N2/3 wafer capacity from TSMC recently
Others	\$4.9	\$4.3	\$18.2	\$21.8	\$25.0	New gaming cycle (PS6/xBox) starts in '27/'28. Client & FPGA steady growth.
Total sales	\$10.3	\$16.3	\$50.6	\$111.3	\$163.5	'27E sales may exceed \$100bn (Street's \$75bn). Our DC sales are \$75bn vs Street's \$57bn.
EPS	\$1.53	\$2.87	\$7.81	\$19.46	\$27.37	

Source: Aletheia/TAG (as of July 2026)

We estimate, to-date, AMD has probably sold out ~2m units of MI450/455/500 GPUs for C2026-27!

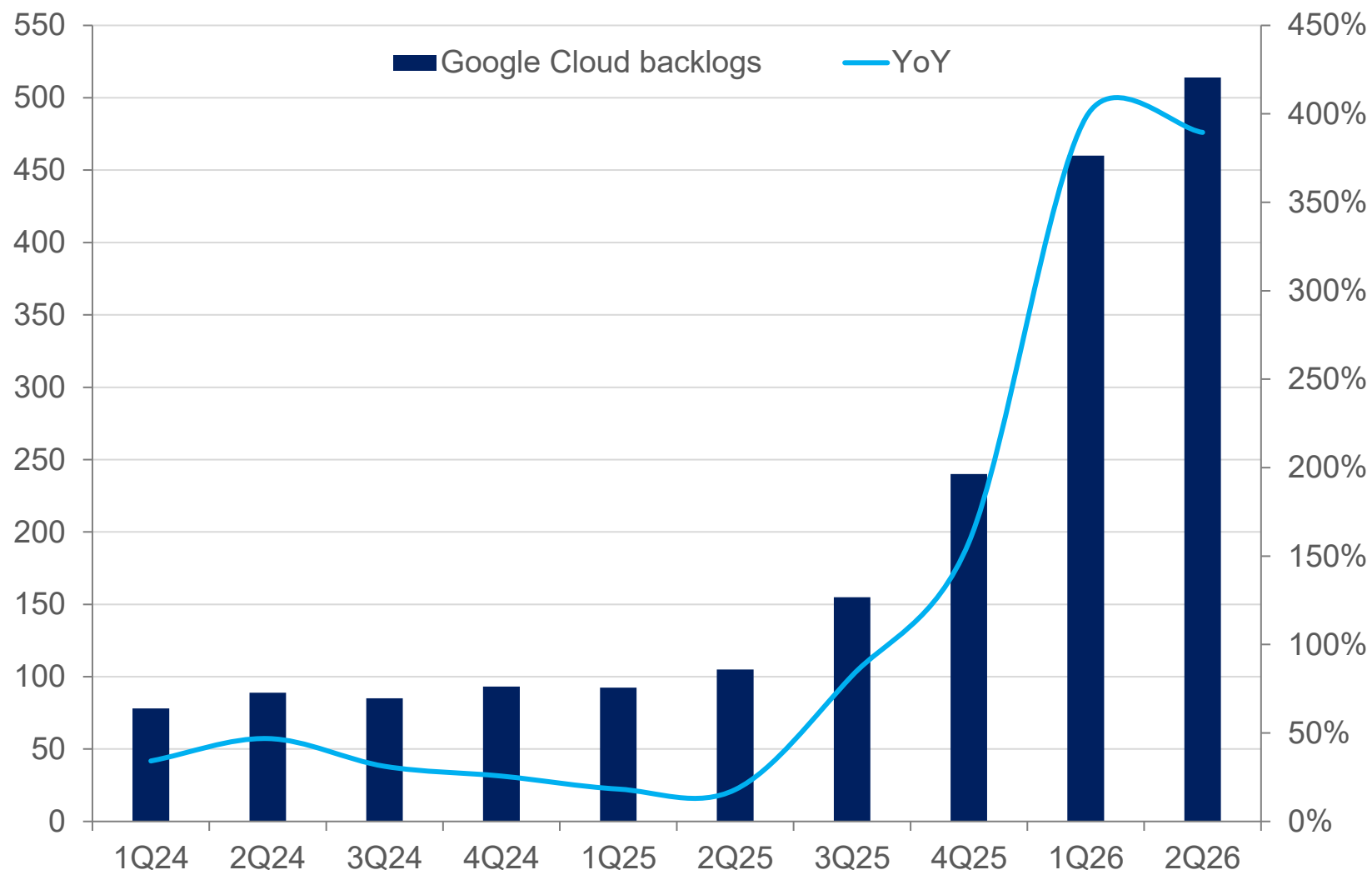
TRACKING INTEL'S PROGRESS

'000 wpm	4Q26E	4Q27E	4Q28E	4Q29E	Remarks
N3	42	65	75	75	The site is @Ireland
18A	45*	52*	80*	85*	Fabs 52+62 to be fully ramped in '28
14A	-	10*	25*	40	Ohio site to resume construction

	Products/customers	HVM shipment	Remarks
Server CPU	Granite Rapids (N3)	Since '25	Expected to be main SKU through '27
	Clearwater Forest (18A+3D)	Mid-26	May be lack of design-wins
	Diamond Rapids (18A+3D)	Mid-27	Delayed twice and a year behind Venice
Client CPU	Panther Lake (18A+TSMC N3+3D)	4Q25/26	Compute tile on 18A, others using TSMC
	Nova Lake (TSMC N2+3D)	4Q26/27	Almost 100% SKU outsourced to TSMC
	Razor Lake (18A+TSMC N2+3D)	4Q27/28	Similar to Panther Lake arrangement
Foundry	Apple's niche or legacy products. Others include AVGO, OAI, GOOG, AWS & MSFT etc	Late-27/28	Needs more 18A-P capacity (in '28) before it can support external customers

*Note: * include pilot line at Oregon site of 15-20k wpm capacity*
Source: Aletheia/TAG (June 2026)

GOOGLE CLOUD PLATFORM BACKLOG ORDERS (USD BN)



Source: Google, Aletheia/TAG (Sep 2026)

TPU DESIGN PARTNERS (AS OF SEP '26)

Where does MRVL fit in?



	TPU V8		TPU V8.5	TPV V9		TPU V10	
Variants	Inference	Training	Inference	Inference	Training	Inference	Training
Compute	AVGO	GOOG	AVGO	GOOG	GOOG	GOOG	GOOG
Logic/Base	-	-	-	MTK	MTK	GOOG	AMD/AVGO
I/O	AVGO	MTK	AVGO	MTK	MTK	MTK	AMD/AVGO
3DIC	-	-	-	-	-	GUC?	AMD/AVGO
HBM procurement	AVGO	AVGO	AVGO	GOOG	GOOG	GOOG	GOOG
HVM	2H26	4Q26	2H27	1Q28	1Q28	1Q29	1Q29

Source: Aletheia/TAG (Sep 2026)

OUR GOOGLE/TPU PORTFOLIO

- Lumentum (optical switches – OCS & EML laser): 30%-40% revenue exposure to Google/TPU
- Celestica (TPU modules and switches): 40% profit
- Isu PetaSys (PCB for TPUs and switches): 50% revenue
- Elite Material (CCL for TPUs and switches): 35%-40% revenue
- Broadcom: GhostFish (V7) and SunFish (V8i) TPU ASICs
- MediaTek: ZebraFish (V8t) and HumuFish (V9i&t) TPU ASICs
- EZconn (optical cables): 85%-90% revenue
- GUC (Axion 2&3 CPUs): 50% revenue
- Others key suppliers: AVC (cold plate) & TTM (PCB)

Note: All sales and profit exposure are based on FY27E estimates

LATEST TRENDS AND UPDATES FOR MEMORY

- Outlook for 2027 HBM pricing negotiations and switch in market share
- Evolution of memory architecture in NVDA's AI hardware
- NAND's role in optimizing memory technology and cost
- Fine-tuning of existing business models:
 - Signing LTAs to lock in prices and volume
 - Memory devices are consigned by customers such as:
 - 1) eSSD in BF4 STX Storage Rack, SoCAAM in Vera CPU rack & standalone Vera CPU, instead of being bundled in the hardware systems like VR NLV;
 - 2) HBMs in TPU V8t, V9 and V10 is procured by Google
- Details of LTA and YTD capital returns

UPDATE ON 2027 HBM ASP NEGOTIATIONS

\$/1GB	CY25	CY26E	CY27E	Comments
SKHY <---> NVDA [HBM3E]	\$16.0	\$14.0	\$18-20	<i>Below expectations but volume is limited</i>
Market price ¹ [HBM3E]	\$21.3	\$16.5	\$22-24	10%-15% below expectations. Remains mainstream with ASICs. AVGO ASP is closer to NVDA range.
SKHY <---> NVDA [HBM4]	NA	\$18.5	\$24-28	<i>>20% below expectations. ASP to be finalized.</i>
Market price ¹ [HBM4]	NA	>\$20	\$32-36	In line with expectations
Market price ¹ HBM4E	NA	NA	\$42-44	In line with expectations but limited volume

Note: 1-Market price refers to all buyers/sellers except transactions between NVDA/SKHY

Source: Aletheia/TAG (Sep 2026)

- ▶ 12-Hi HBM stack is the mainstream in 2027 (Rubin, MI500 & TPU V8) but...
- ▶ ... Rubin Ultra and TPU V9 revert back to 8-Hi in 2028...
- ▶ ...16-Hi looks a distance away

2027 HBM VOLUME ALLOCATIONS

GPU/ASIC	Samsung	SKHY	Micron	Comments
NVDA	30%	>50%	<20%	SKHY remains the No1 suppliers to NVDA
AMD	70%	20%	10%	SEC enjoys closed relationships with AMD
TPU V8i (AVGO)	100%	-	-	SEC enjoys closed relationships with AVGO/GOOG
TPU V8t (MTK)	20%	40%	40%	
AWS	>50%	<30%	20%	SEC remains the No1 supplier to AWS

Source: Aletheia/TAG (Sep 2026)

- Expect Samsung to match/overtake SKHY to become No1 in HBM market in 2027
- Samsung is capturing the majority HBM share of the fastest-growing AI devices – TPU V8 (60% share) and MI455/500 (70%)
- SKHY's steep ASP discount to NVDA places itself at a disadvantage

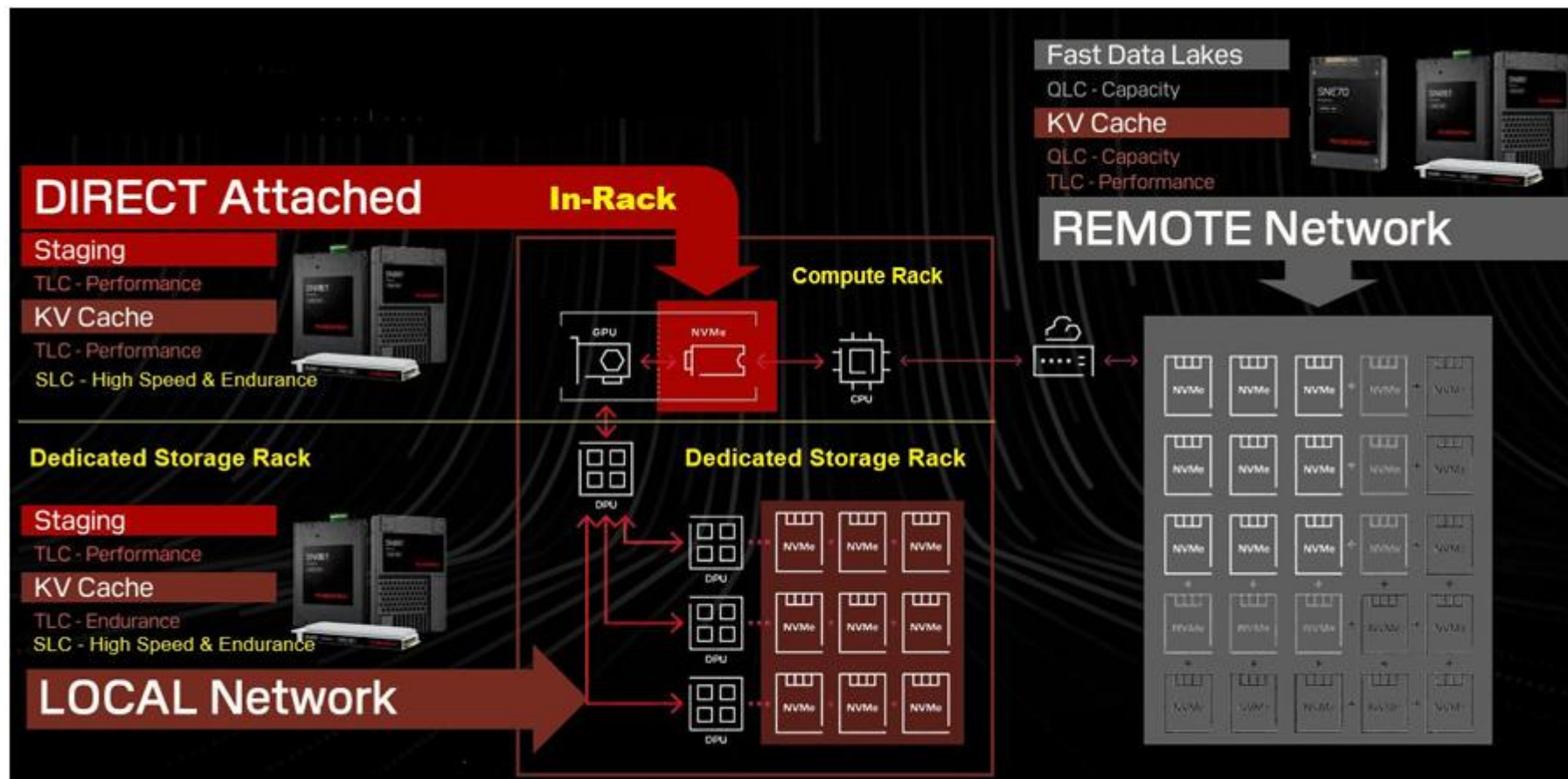
ROADMAP OF TPU's & AMD's HBM SPECS

HVM schedule	TPU7	TPU8 (4Q26)		TPU V9 (1Q28)		TPU V10 (1Q29)	
Variants	-	Inference	Training	Inference	Training	Inference	Training
Types	3E	3E	3E	4E	4E	4E	4E
Layers	8-Hi	12-Hi	12-Hi	8-Hi	8-Hi	8-Hi	8Hi
Stackings	8x	8x	6x	12x	12x	16x	16x
Total density	192GB	288GB	216GB	384GB	384GB	512GB	512GB
Total bandwidth	7.4TB/s	8.6TB/s	6.5TB/s	~40TB/s	~40TB/s	~40TB/s	~60TB/s

HVM schedule	MI355X	MI455X (3Q26)	MI550X (4Q27)		MI600 (4Q28)		
Types	3E	4	4E	4E	4E/5	4E/5	4E/5
Layers	12-Hi	12-Hi	8-Hi	12-Hi	N.A.	N.A.	N.A.
Stackings	8x	12x ¹	16x (quad-die) ¹	8x (dual-die) ¹	N.A.	N.A.	N.A.
Total density	288GB	432GB	512GB	384GB	N.A.	N.A.	N.A.
Total bandwidth	7TB/s	23.3TB/s	~65TB/s	~33TB/s	N.A.	N.A.	N.A.

Note: 1-MI455X and MI550X (dual-die) support Helios single rack while MI550X (quad-die) is used for pod structure (512 GPU)
Source: Aletheia/TAG (Sep 2026)

EVOLUTION OF AI MEMORY ARCHITECTURE



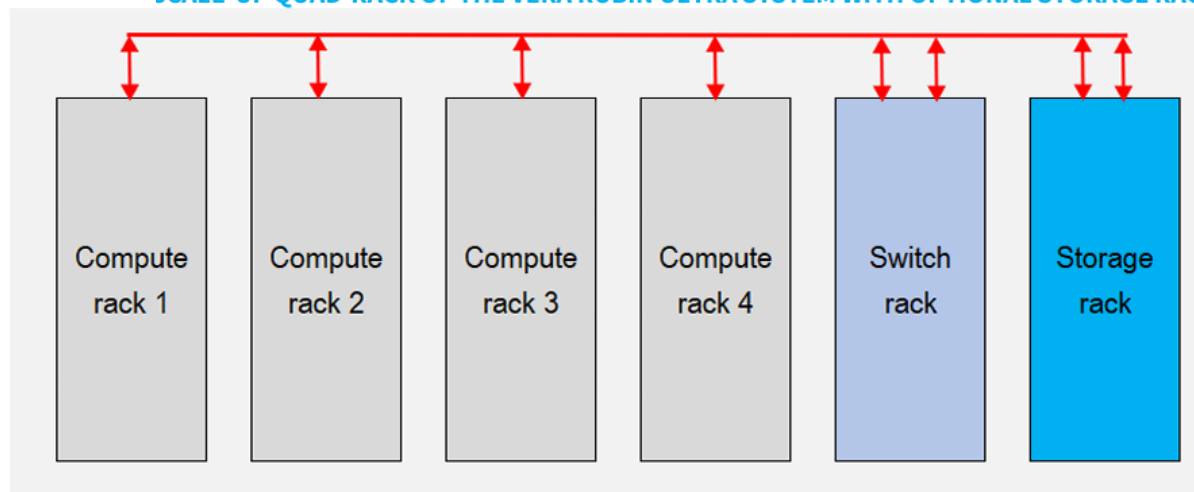
Source: SNDK Investor Day (August 2026), Aletheia/TAG (with additional comments in Yellow)

EVOLUTION OF AI MEMORY ARCHITECTURE

Memory Architecture	Grace Blackwell		Vera Rubin	
	GB200	GB300	VR200	VR300
HBM	13.8TB (3E)	20.7TB (3E)	20.7TB (4)	18.3TB (4E)
SoCAAM/DRAM	18.4TB	18.4TB	27.6TB	27.6TB
Local/direct attached eSSD	-	288TB	576TB	TBD
Total memory density in a compute rack	32.2TB	327.1TB	624.3TB	TBD
Dedicated network storage eSSD	-	-	1,152TB	TBD

Source: NVDA & Aletheia/TAG (August 2026)

SCALE-UP QUAD-RACK OF THE VERA RUBIN ULTRA SYSTEM WITH OPTIONAL STORAGE RACK



Each Rubin will have up to 24.7TB of memory, up 6x vs the GB300. Expect more in the Rubin Ultra.

Source: Aletheia/TAG (August 2026)

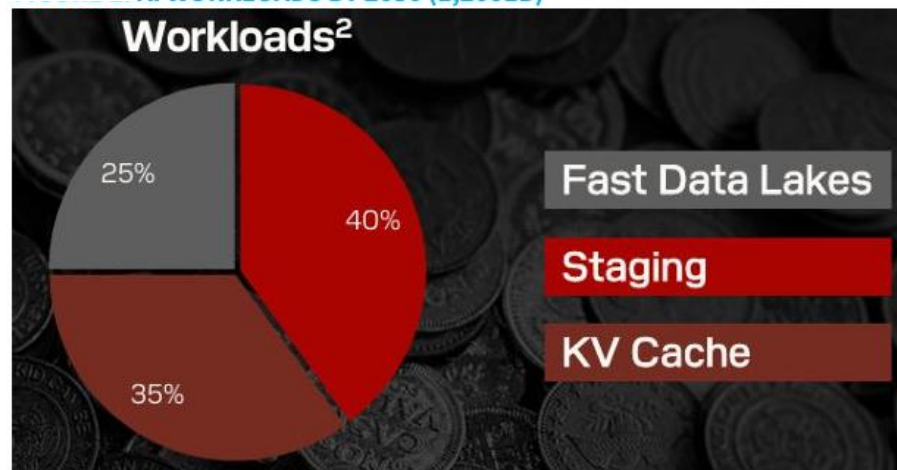
EVOLUTION OF AI MEMORY ARCHITECTURE

SPECIFICATIONS OF NAND USED IN AI DATACENTER COMPARED TO SMARTPHONE

Key specifications	KV cache ¹	Staging	Fast data	Smartphone
Flash type	SLC	TLC	QLC	QLC (USF4)
Write/read latency	4-5/3-5μs	15-25/50-70μs	1,000-2,000/75-110μs	1,000-2,500/70-120μs
Write/read speed	NA/10MIOPS	Up to 1.8/5.5MIOPS	1MIOPS	0.2MIOPS
Write Bandwidth	23GBps@PCIe6	14-17GBps@PCIe6	7GBps ⁵	Up to 0.40GBps ⁵
Endurance ²	Up to 50DWPD	Up to 5DWPD	Up to 0.7DWPD	Up to 0.3DWPD
ASP ³ per GB	TBD ⁴	\$0.45-0.50	\$0.35-0.40	\$0.15-0.25

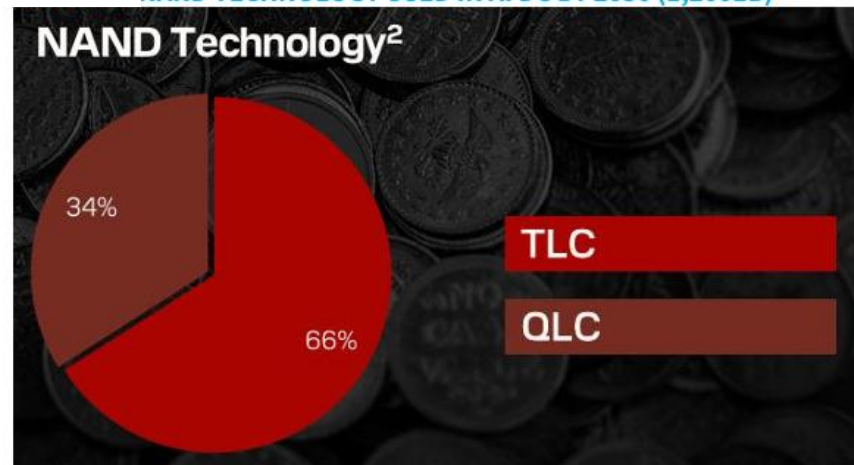
Note: 1-these are based on Kioxia's XL-Flash specification. 2- Daily Write Per Day i.e., 10DWPD means write 10,000GB of data per day for 5 years period. 3-based on the estimated contract price in 3Q26. 4-expectation is that XL-Flash will be prices at 6-10x that of high-performance TLC NAND. 5-the read bandwidth for QLC for fast data in AI DC is 15GBps and smartphone QLC is <5GBps.
Source: Aletheia/TAG (as of August 2026)

AI WORKLOADS BY 2030 (1,200EB)



Source: SNDK Investor Day (August 2026)

NAND TECHNOLOGY USED IN AI DC BY 2030 (1,200EB)



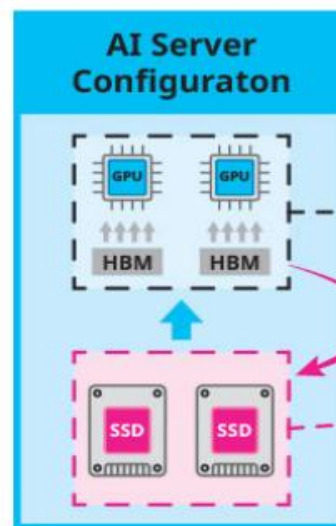
Source: SNDK Investor Day (August 2026)

EVOLUTION OF AI MEMORY ARCHITECTURE

KIOXIA Path to 100 MIOPS



Source: Kioxia Technology Blog (April 2026)



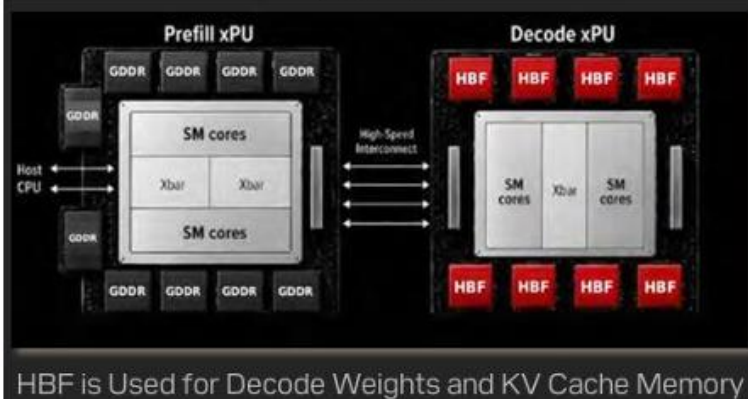
NVDA+Kioxia opt for High-Speed NAND

HBM capacity is fixed to a GPU and cannot scale

AI workload growth supported by SSDs

Super High IOPS SSD enables AI scalability and performance without HBM expansion

Disaggregated



Source: SNDK IR day (Aug 2026)

HBF (+HBM)

(N) stack positions with (k) HBFs and (N-k) HBMs



HBM/HBF Cached



SKHY+SNDK push for HBF

LTAs FOR MEMORY (AS OF 2Q26)

	Samsung	SKHY³	Micron	Kioxia	SNDK
Duration	5 years	5 years	3-5 years ⁴	3+years	4 years
Targeted coverage	60-70%	50-60%	20-33% ⁵	~50% in 2H26	2/3 in 27/28
Customers	5+5 ¹	10	16	3	8
Ceiling price	Yes	No ³	Yes	No	Yes
Floor price	Yes	Yes	Yes	Yes	Yes
Deposit/ prepayment	Yes ²	No	\$22bn	No	\$16.5bn
Total value	N.A.	N.A.	\$100bn	N.A.	\$94bn

*Note: 1- 5 AI customers confirmed as of 2Q26. Five more clients to be confirmed in 3Q26. 2-Cash payment is 25% of value of contract (covering DRAM& NAND). 3-SKHY's LTA doesn't applied to NVDA as the HBM ASP is determined annually and server DRAM (SoCAAM) is semi-annual or quarterly. ASP offered to NVDA will be at a discounted to market price. 4-Auto customers signed 3 year contract while others are on five years term. 5-20% for DRAM and 33% for NAND by 2030.
Source: Aletheia/TAG (Sep 2026)*

LTAs = CAPITAL RETURNS ARE HAPPENING

FCF (bn) 21 Aug '26	Mktcap	CY25	CY26E	CY27E	Total FCF	FCF yield
Samsung	\$1,260	\$26.6	\$178.3	\$313.8	\$518.7	41.2%
SKHY	\$930	\$24.3	\$113.3	\$214.4	\$352.7	37.9%
Micron	\$1,100	\$20.6	\$89.8	\$175.4	\$285.8	26.0%
Kioxia	\$188	\$1.8	\$24.1	\$66.6	\$92.6	49.3%
SNDK	\$234	\$5.6	\$20.0	\$35.0	\$60.6	25.9%
Total	\$3,712	\$79.0	\$425.5	\$805.3	\$1,310	35.3%

Source: Aletheia/TAG (Sep 2026)

- ▶ On 19 August, SKHY announced \$29bn of stock buybacks by 18 November
- ▶ On 21 August, Samsung approved up to \$80bn of capital returns for CY26
- ▶ Kioxia and SNDK bought back ~\$10bn of stock in April-August; ~100% of FCF
- ▶ Micron will be next once the US Chip Act lockup expires on 9 December

SEMI CAPEX OUTLOOK

- Spending on WFE and advanced packaging could accelerate in '27 and grow further in '28 on:
 - ✓ Readiness of new cleanrooms globally
 - ✓ TSMC repurposing 200mm fabs for adv packaging, initiating PLP investments and quadrupling SolC capacity in two years
 - ✓ Up to 6/6-8 cleanrooms ready for memory expansion in 27E/28E
 - ✓ Potential price hike up to 10% in CY1Q27
- TSMC is set to spend as much as \$250bn on capex in '26E-'28E
- SK Hynix/MU to spend over \$150bn EACH in '26E-'28E
- Intel needs to step-up capex aggressively in '27-'28E
- Four HBM packaging fabs and up to 10 adv logic packaging sites to ramp production from '26-'28E,

DOES ASML HAVE SUFFICIENT EUV CAPACITY?

Estimated Low NA EUV scanner '**demand**' by customer (excluding Terafab)

Customers	2025	2026E	2027E	2028E
TSMC	18	24	45	50 ↑
Samsung	5 (5+0)	18 (12+6)	24 (22+2) ↑	20 (18+2)
SK Hynix	8	14	20	24 ↑
Intel	10	6	11	20 ↑
Micron	2	1	3	6
Rapidus	1	1	1	2
Nanya Tech	-	-	-	2
Total	44	62	104	124

Note: 2025 figures are installed EUV quantity. 2026E and onward are customers' demand. All figures are Low NA EUV excluding Hi NA EUV.

Source: Aletheia/TAG (June 2026)

**ASML's 30% capacity growth in C27/28
won't be able to meet demand**

GLOBAL CLEANROOM READINESS & TOOL MOVE-IN SCHEDULES

Logic	Site	C2025	C1H26	C2H26	C1H27	C2H27	C2028/'29
TSMC	Fab18	-	-	-	P9 (N3)	P10 (N3/A14)	P11 (N3/A14)
	Fab20	P1 (N2/A16)	P2 (N2/A16)	P3 (A14^)	P3 (A14)	-	P4 (A14)
	Fab22	P1 (N2/A16)	P2 (N2/A16)	-	P3 (N2/A16)	P4 (N2/A16)	P5 (N2/A16)
	Fab25	-	-	-	-	P1 (A14)	P2 (A14)/ P3-4
TSMC-AZ	Fab21	P1AB (N4)	-	P2A (N3)	P2B (N2)	-	P3A (N2)/ P3B
TSMC-JP	Fab23	-	-	-	-	P2 (N3)	-
TSMC-DE	Fab24	-	-	-	-	-	P1 (12/22nm)
Intel	Fab62	-	-	-	-	18A	
	Fab27 ¹	-	-	-	-	-	P1 (14A)
Samsung	Taylor	-	-	P1			P2
DRAM	Site	C2025	C1H26	C2H26	C1H27	C2H27	C2028
Samsung	PK4	-	P1 (HBM)		P2 (HBM)		
	PK5	-	-	-	-	-	P1
SK Hynix	M15X	M15X (HBM)					
	Yongin	-	-	F1P1 (HBM)		F1P2	F1P3
Micron	Boise	-	-	-	P1		P2
	Japan	-	-	-	-	P2 (HBM)	
	Maioli	-			P5-1 (DDR5/HBM)		P5-2 (DDR5/HBM)
Nanya	Taishan	-	-	P1 (DDR4/5)			P2 (DDR5)
Winbond	Kaohsiung	-	KS (DDR4)				
3D NAND	Site	C2025	C1H26	C2H26	C1H27	C2H27	C2028
Kioxia	Yokkaichi	Fab7					
	Kitakami	Fab2					Fab3 in '29
Samsung	PK5	-	-	-	-	-	P1 or P2
Hynix	Dalian	Fab2	-	-	Fab2		
	Cheongju	M17					4Q28
Micron	Singapore	-	-	-	-	-	P2

Note: fabs in Red are the latest revision.

1-Intel may resume Fab27 site at Ohio.

Source: Aletheia/TAG (Sep 2026)

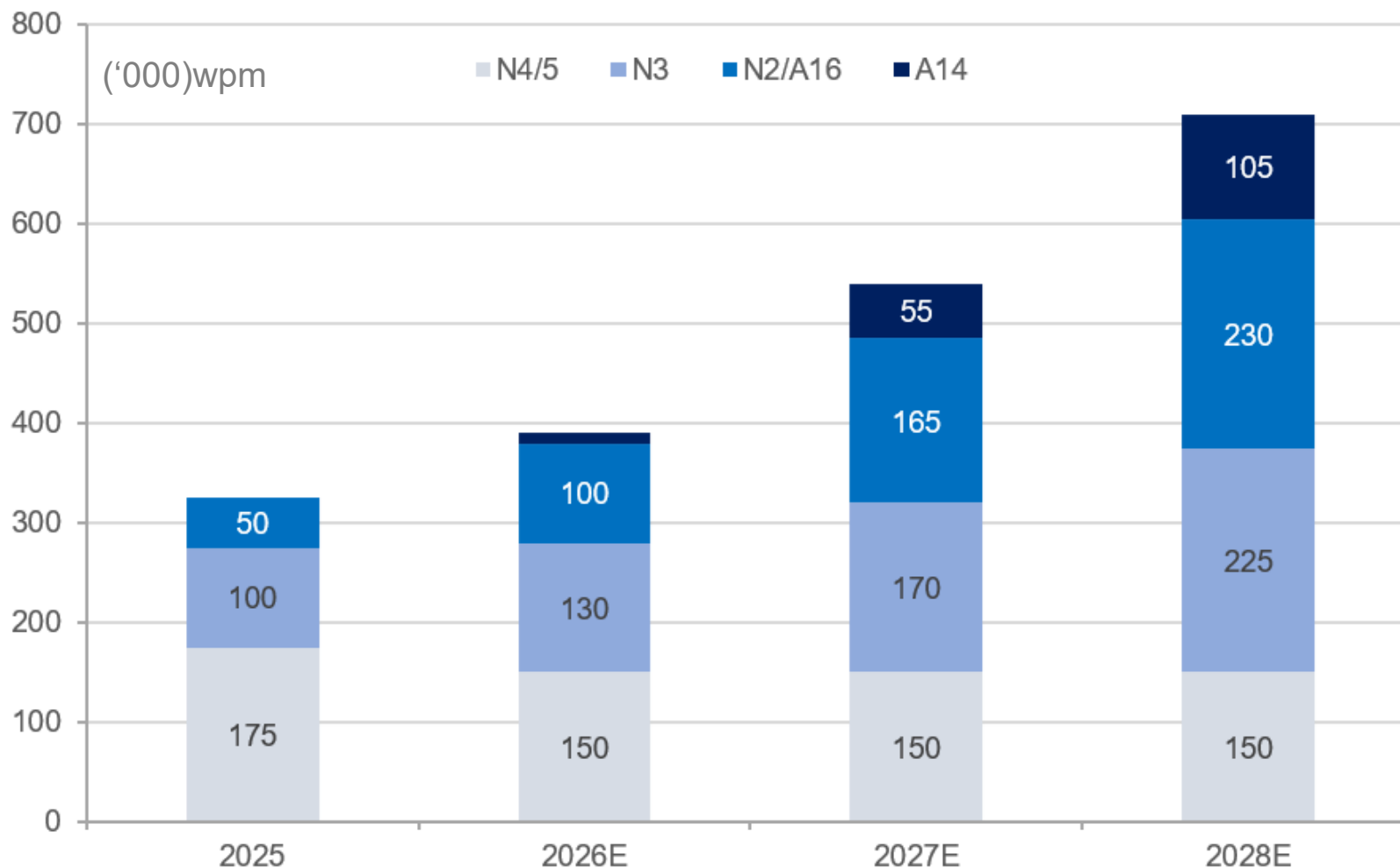
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TSMC'S MID-TERM CAPACITY EXPANSION PLAN

Node	Sites	Name	2025	2026	2027	2028
N3	TN	Fab18	-	-	P9	P10+ (P11)
	Capacity		-	-	25	30
N2/A16	Sites	Name	2025	2026	2027	2028
	HS	Fab20	P1	P2	-	-
	KS	Fab22	P1	P2	P3+P4	P5
	TN	Fab18			P10+ (P11)	
	Capacity		50	50	50	55
A14	Sites	Name	2025	2026	2027	2028
	HS	Fab20	-	P3 (pilot)	P3	P4
	TC	Fab25	-	-	P1	P2
	Capacity			10	45	50
	Total capacity in TWN		50	60	120	135
N2/3/4 N6/7/3	Sites	Name	2025	2026	2027	2028
	Arizona	Fab21	P1AB	P2A	P2B	P3A
	Capacity		25	10	15	10
	Kumamoto	Fab23	-	-	P2A	P2B
	Capacity		-	-	15	25
	Total WW capacity		75	70	150	170

Source: Aletheia/TAG (as of June 2026)

TSMC'S MID-TERM CAPACITY EXPANSION PLAN



Source: Aletheia/TAG (as of June 2026)

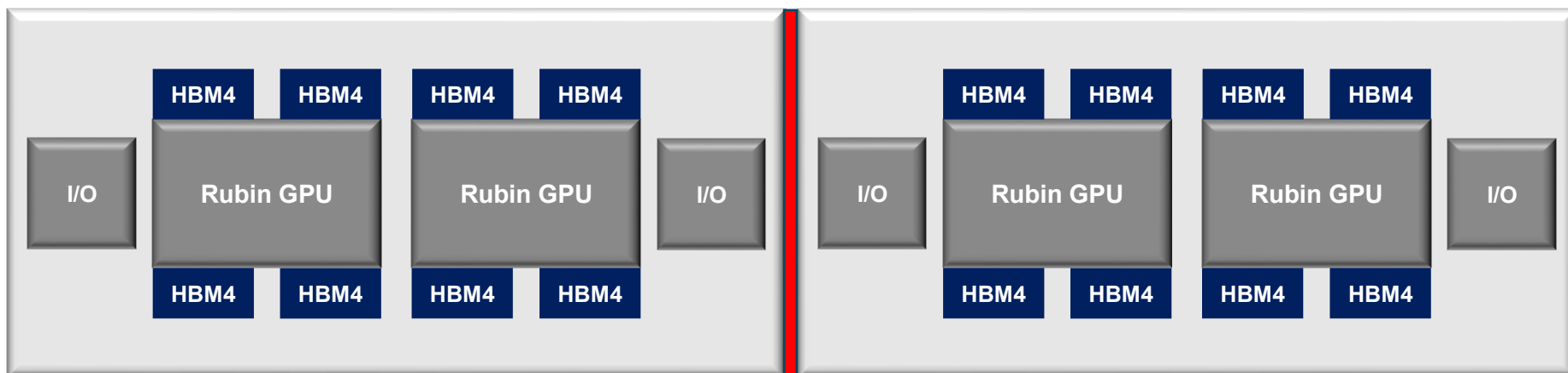
ESTIMATED TSMC's ADV. PACKAGING CAPACITY

Year-end capacity ('000 wafers/month)	2024	2025	2026E	2027E	2028E
CoWoS	35	70	120	180	180
AP 1/2/3/5/6	35	60	80	80	80
AP7 (P1/P2)			-	20	20
AP8 (P1/P2)		10	40	80	80
CoPoS	-	-	1	8	23
AP7 (P4) & pilot			1	8	23
SoIC	8	10	20	32	60
AP 1/2/3/5/6	8	10	20	20	20
AP7 (P2/3)			-	12	40
InFO	100	90	50	20	-
AP 1/2/3/5/6	100	90	50	20	-
WMCM	-	10	75	125	145
AP 1/2/3/5/6		10	50	80	100
AP7 (P1)			25	45	45

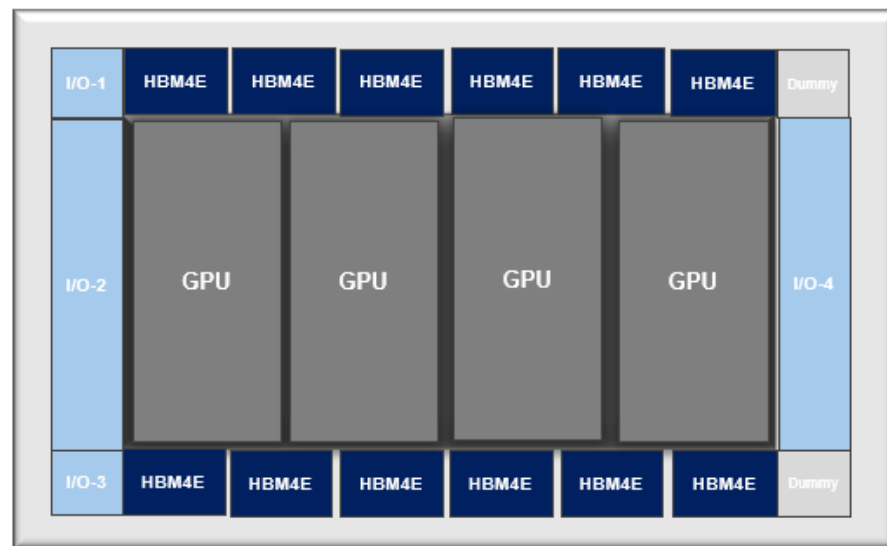
Source: Aletheia/TAG (as of June 2026)

TSMC is likely to initiate commercialization of CoPoS in late-27 with the legacy 200mm Fab5 in Hsinchu

TOPOLOGY OF NVDA's GPUS



Source: Aletheia/TAG (Sep 2026)



Source: Aletheia/TAG (Sep 2026)

2H27 nVidia's Rubin Ultra GPU (Quad-die):

- ▶ Similar Rubin (N3+N4 & HBM4/4E)
- ▶ WoS stitching post CoW

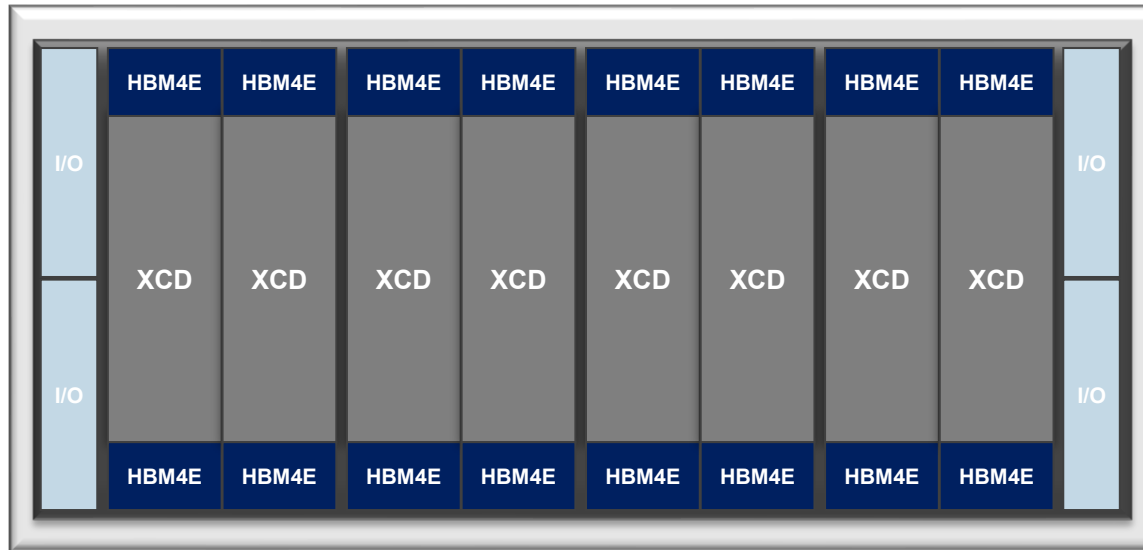
2H28 nVidia's Feynman GPU:

- ▶ 4x GPU tiles on A16 BSPD
- ▶ 2x base tiles on A16 BSPD
- ▶ 4x types of I/O tiles on N3P
- ▶ 12x HBM4E (8/12Hi)
- ▶ Estimated reticle size 6-7x
- ▶ CoWoS or CoPoS + SoIC

TOPOLOGY OF AMD's GPUs

MI500 4Q27/28 GPU:

- ▶ 9.5x reticle size (Quad-die)
- ▶ 8x N2P GPU dies
- ▶ 4x N3P based die
- ▶ 4x N3P I/O
- ▶ 16x HBM4E (12Hi)
- ▶ 3DIC



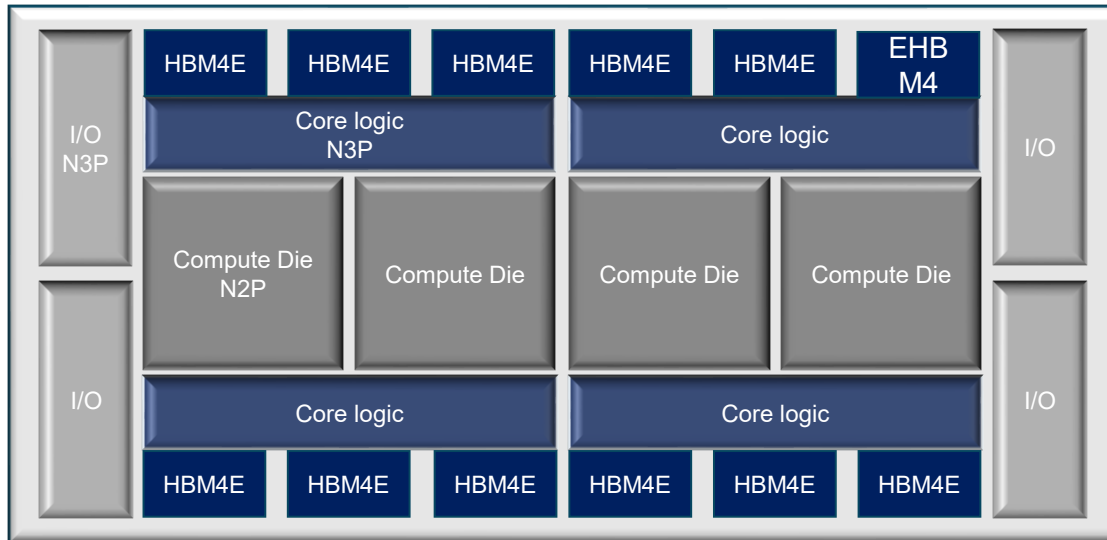
Source: Aletheia/TAG (Sep 2026)

MI600 4Q28/29 GPU:

- ▶ As large as 15x reticle size
- ▶ N2P/A14
- ▶ HBM4E/5
- ▶ 3DIC with SRAM cache

**There could be
multiple variants**

TOPOLOGY OF GOOGLE'S TPUS



Source: Aletheia/TAG (Sep 2026)

HumuFish 1H28 TPU (V9i&t):

- ▶ 9.5x reticle size
- ▶ 4xN2P compute dies
- ▶ 4x N3P logic dies
- ▶ 4x N3P I/Os
- ▶ 12xHBM4E (8Hi)
- ▶ EMIB-T/panel packaging

**Very complicated
& super powerful
(~12KW TDP)**

TPU (V10i) 1H29:

- ▶ ~15x reticle size
- ▶ 16xN2P compute dies
- ▶ 8x N2P base dies
- ▶ 6x N2P I/Os
- ▶ 16xHBM4E (8Hi)
- ▶ 3DIC + EMIB-T/panel packaging

Source: Aletheia/TAG (Sep 2026)

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